DOPPLER RADAR SYSTEM

A Project Report

submitted to the *APJ Abdul Kalam Technological University*

in partial fulfillment of the requirements for the award of degree

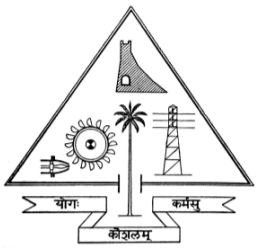
Bachelor of Technology

in

Electronics and Communication Engineering

by

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OCTOBER 2026

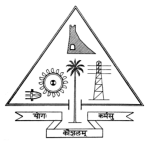
DECLARATION

We hereby declare that the subject project entitled "Doppler Raddar System", submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology at APJ Abdul Kalam Technological University, Kerala, is a bona fide work carried out by us under the supervision of Assistant Prof. Deepak S. This project sents our original work. Wherever the ideas or words of others have been used, the original sources have been properly cited and referenced. We have adhered to the principles of academic honesty and integrity and have not fabricated, misrepresented, or plagiarized any data, ideas, or facts. This project has not previously been submitted, in whole or in part, for the award of any degree, diploma, or similar title at any other university or institution.

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CERTIFICATE

This is to certify that the subject project report entitled **Real-Time Audio Synthesizer** submitted by

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to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the B.Tech. degree in Electronics and Communication Engineering is a bona fide record of the subject project carried out by them under our guidance and supervision. This project report, in any form, has not been submitted to any other University or Institute for any purpose.

Project Guide	Project Coordinator
Assistant Prof. Deepak S	Assistant Prof. Athira V E

ACKNOWLEDGEMENT

We express our sincere gratitude to the faculty members of the Department of Electronics and Communication Engineering, Government Engineering College, Thrissur, for their academic guidance and support during this project. We thank our project guide and project coordinator for their suggestions, and the Head of the Department for providing an environment that supports technical learning. We also acknowledge the technical references used to understand Doppler radar and digital signal processing. Finally, we thank our classmates and families for their encouragement.

ABSTRACT

This project presents a software simulation of vehicle speed estimation using the Doppler effect and digital signal processing. A continuous-wave Doppler signal is generated synthetically, noise is added, and a finite impulse response low-pass filter is applied before frequency-domain analysis. The Fast Fourier Transform is used to identify the dominant Doppler component. The detected frequency is converted into radial velocity, after which a known radar deployment angle is used to estimate the vehicle-path speed. The simulation also displays time-domain signals, a frequency spectrum, speed comparison, angle effects, and threshold-based warning levels. The work demonstrates the signal-processing sequence in a controlled and repeatable environment. Since only simulation has been implemented, the reported results are outputs of synthetic data and do not establish the accuracy of a physical radar device.

Keywords: Doppler radar, vehicle speed estimation, digital signal processing, FIR filter, FFT, spectral peak detection, angle correction.

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CHAPTER 1

INTRODUCTION

Digital Signal Processing (DSP) is used to analyse sampled signals and extract useful information from them. In radar applications, the received signal contains information about the movement of a target. One important application is vehicle speed estimation, where the Doppler effect relates the frequency shift of a reflected radio signal to the motion of the vehicle. Radar-based estimation is contactless and can be studied without installing a sensor on the vehicle itself.

A continuous-wave (CW) Doppler radar transmits a signal at a fixed carrier frequency. When the signal reflects from a moving object, the frequency of the received signal differs from the transmitted frequency. For a monostatic radar, where transmission and reception occur at approximately the same location, the signal travels to the target and back. This two-way path introduces a factor of two in the Doppler-frequency relationship. The measured frequency shift represents the radial component of the target velocity, which depends on the angle between the radar line of sight and the direction of travel.

The project implements these concepts in a software simulation rather than a physical radar setup. A synthetic Doppler signal is generated from the selected simulation parameters. Noise is introduced to represent disturbance, and a finite impulse response (FIR) low-pass filter is applied to prepare the signal for spectral analysis. The Fast Fourier Transform (FFT) converts the sampled signal into a frequency-domain representation, allowing the dominant Doppler peak to be detected. The peak frequency is then converted to radial speed and corrected using a specified radar angle.

The simulation interface provides adjustable parameters and visual outputs, making it possible to observe how signal noise, frequency detection, and radar geometry influence the estimate. The project therefore serves as a demonstration of a complete DSP workflow, from signal generation to speed estimation and warning classification. Its conclusions are limited to the simulated conditions because no physical radar measurements or hardware validation were performed.

1.1 Objectives

- To develop a software simulation for vehicle speed estimation using the CW Doppler principle.
- To generate a synthetic Doppler signal and model noise at a selected signal-to-noise ratio.
- To apply FIR low-pass filtering before frequency-domain analysis.
- To detect the dominant Doppler frequency using FFT-based spectral peak detection.
- To convert the detected frequency into radial velocity and apply known-angle correction.
- To display simulation parameters, plots, estimated speed, and warning classification.

1.2 Problem Statement

A Doppler speed-estimation algorithm must identify a frequency shift from a sampled signal that may contain noise and unwanted components. The estimated frequency must then be converted correctly into radial velocity, while the radar angle must be considered to estimate the speed along the vehicle path. This project addresses these processing steps through a reproducible simulation. It does not claim to validate the performance of a real-world radar sensor.

1.3 Scope

The work is limited to a single-channel CW Doppler simulation with a dominant target component. The implemented scope includes synthetic signal generation, noise addition, FIR filtering, FFT peak detection, radial-speed calculation, angle correction, and threshold-based warning display. Physical sensor acquisition, analogue signal conditioning, hardware ADC interfacing, multi-target tracking, and field testing are outside the scope of the implemented project.

CHAPTER 2

EXISTING AND PROPOSED SYSTEM

2.1 Existing System

Vehicle speed can be estimated using road-embedded detectors, optical sensors, camera-based tracking, and radar instruments. Embedded detectors infer vehicle passage or speed from changes in electrical or magnetic responses. Optical systems use timing or position changes, while camera-based systems track vehicles across successive frames. Radar systems estimate speed from the Doppler shift of reflected radio waves. These methods differ in cost, installation requirements, sensitivity to environmental conditions, and processing complexity.

2.2 Proposed System

The proposed work is a DSP simulation of a CW Doppler speed estimator. The program generates a synthetic signal associated with a selected target speed, introduces noise according to a specified signal-to-noise ratio, and applies a Hamming-window FIR low-pass filter. An FFT is then computed, and the dominant peak within the configured Doppler range is selected. The detected frequency is converted to radial velocity using the radar carrier frequency and the speed of light. A known deployment angle is used for cosine correction, and the resulting estimate is compared with configurable warning thresholds.

2.3 Project Contribution

The main contribution is the integration of the signal-processing stages into a visual and repeatable simulation. Parameters such as speed, radar angle, noise level, and warning thresholds can be changed to examine their effect on the output. The interface presents numerical results alongside plots of the time-domain signal, FFT spectrum, speed comparison, angle effect, and warning level. This contribution is educational and algorithmic; it is not a hardware implementation.

2.4 Implementation Boundary

All reported numerical results originate from synthetic data generated by the simulation. Therefore, the error shown by the interface indicates agreement with the selected simulated input, not independently measured accuracy. A physical implementation would require a radar sensor, analogue front-end, sampling hardware, calibration, and comparison with a trusted speed reference.

CHAPTER 3

GENERAL ARCHITECTURE

3.1 System Architecture

Simulation parameter selection → Doppler signal generation → Noise addition → Hamming-window FIR filtering → FFT magnitude spectrum → Doppler peak detection → Radial-speed calculation → Angle correction → Warning classification

Figure 3.1: Processing flow of the Doppler speed-estimation simulation.

3.2 Description of Each Block

Parameter selection: The user selects the simulated target speed, radar angle, signal-to-noise ratio, random seed, and warning thresholds. These values establish the conditions for a simulation run.

Doppler signal generation: A sinusoidal intermediate-frequency signal is generated from the theoretical Doppler frequency associated with the selected radial velocity. The radar carrier frequency is used in the conversion between frequency and velocity.

Noise addition: Random noise is added to represent disturbance in the signal. The selected signal-to-noise ratio controls the relative signal and noise power, while a fixed random seed allows repeatable runs.

FIR filtering: A Hamming-window FIR low-pass filter reduces unwanted components before spectral analysis. Filter order and cutoff frequency affect noise suppression and the signal response.

FFT and peak detection: The filtered samples are transformed into the frequency domain. The dominant spectral peak in the configured frequency range is selected as the Doppler-frequency estimate.

Speed calculation and warning: The detected frequency is converted into radial speed. The specified radar angle is then used for cosine correction, and the estimated speed is compared against the selected warning thresholds.

3.3 Simulation Software and Main Parameters

The project is implemented as a software-based simulation with an interactive interface. The signal-processing sequence uses a synthetic input, noise generation, FIR filtering, FFT analysis, and numerical speed conversion. The parameters used for the documented observation are presented in Chapter 6 so that the procedure remains separate from the particular test case.

CHAPTER 4
METHODOLOGY

4.1 Doppler Principle

For a monostatic continuous-wave radar, the Doppler frequency shift is related to radial velocity by

$$f_d = \frac{2v_r f_c}{c} \tag{4.1}$$

where f_d is the Doppler frequency shift in hertz, v_r is radial velocity in metres per second, f_c is the radar carrier frequency, and c is the speed of light. Rearranging the expression gives

$$v_r = \frac{f_d c}{2f_c}. \tag{4.2}$$

The factor of two represents the outgoing and reflected paths. The result is radial velocity rather than necessarily the full speed along the road.

4.2 Radar Angle Correction

When the radar line of sight forms an angle θ with the direction of travel, the radial velocity is

$$v_r = v\cos\theta. \tag{4.3}$$

If the angle is known, the vehicle-path speed can be estimated using

$$v = \frac{v_r}{\cos\theta}. \tag{4.4}$$

The correction compensates for the fact that only the component of velocity along the radar line of sight contributes to the Doppler shift. The angle must be supplied or estimated independently; a single-channel CW Doppler signal does not determine it by itself. An inaccurate angle produces a corresponding bias in the corrected speed.

4.3 Sampling and FFT Analysis

The continuous signal is represented by discrete samples at sampling frequency f_s. According to the Nyquist criterion, the sampling frequency must be greater than twice the highest frequency component that is to be measured. The FFT transforms a finite sample record from the time domain to the frequency domain. The magnitude spectrum is examined to locate the dominant Doppler component.
The nominal FFT-bin spacing is

$$\Delta f = \frac{f_s}{N_{\mathrm{FFT}}}. \tag{4.5}$$

A larger FFT length gives a finer displayed frequency grid, but zero-padding alone does not add new information to the sampled signal. The observation duration, signal-to-noise ratio, and spectral peak-selection method also influence the frequency estimate.

4.4 FIR Filtering and Windowing

The simulation uses a Hamming-window FIR low-pass filter. A finite impulse response filter uses a finite set of coefficients and does not rely on feedback. Windowing shapes the

filter design to manage the transition band and sidelobe behaviour. Filtering can reduce unwanted signal components before the FFT, but the cutoff frequency must preserve the Doppler frequencies of interest.

4.5 Speed Warning Classification

The interface compares the corrected speed estimate with user-configured thresholds and displays a warning category. These thresholds are demonstration settings for the simulation and should not be interpreted as official traffic regulations or a certified enforcement system.

CHAPTER 5
SIMULATION PROCEDURE

The simulation was operated through the following general procedure. The steps describe the implemented workflow without tying the procedure to a single numerical example.

1. Open the Doppler radar speed-estimation simulation interface.
2. Choose the simulated target speed, radar angle, signal-to-noise ratio, random seed, and warning thresholds.
3. Generate the synthetic Doppler signal using the selected simulation parameters.
4. Add noise to the signal according to the specified signal-to-noise ratio.
5. Apply the Hamming-window FIR low-pass filter to the noisy signal.
6. Compute the FFT magnitude spectrum from the filtered samples.
7. Detect the dominant Doppler peak within the configured frequency range.
8. Convert the detected frequency into radial velocity using the Doppler equation.
9. Apply cosine correction using the selected radar deployment angle.
10. Compare the corrected speed estimate with the configured warning thresholds.
11. Record the numerical output and inspect the time-domain, spectrum, speed-comparison, angle-effect, and warning plots.

The parameter values and numerical observations from the supplied simulation run are documented in Chapter 6.

CHAPTER 6
RESULTS AND OBSERVATIONS

6.1 Simulation Parameters

The following values are transcribed from the supplied simulation output and define the observation discussed in this chapter.

Parameter	**Observed configuration**
Selected vehicle speed	82 km/h
Radar angle	30°
Radar carrier frequency	10.525 GHz
Speed of light	3×10^8 m/s
Sampling frequency	10,000 Hz
Simulation duration	2 s
Number of samples	20,000
FFT length	16,384
Signal-to-noise ratio	10 dB
FIR filter order	100
Window	Hamming
Noise random seed	0
Theoretical Doppler frequency	1384.117 Hz
Detected Doppler frequency	1384.277 Hz
Estimated vehicle speed	82.01 km/h
Reported absolute error	0.01 km/h
Reported percentage error	0.01%
Warning output	RED – DANGEROUS OVERSPEED

6.2 Simulation Dashboard

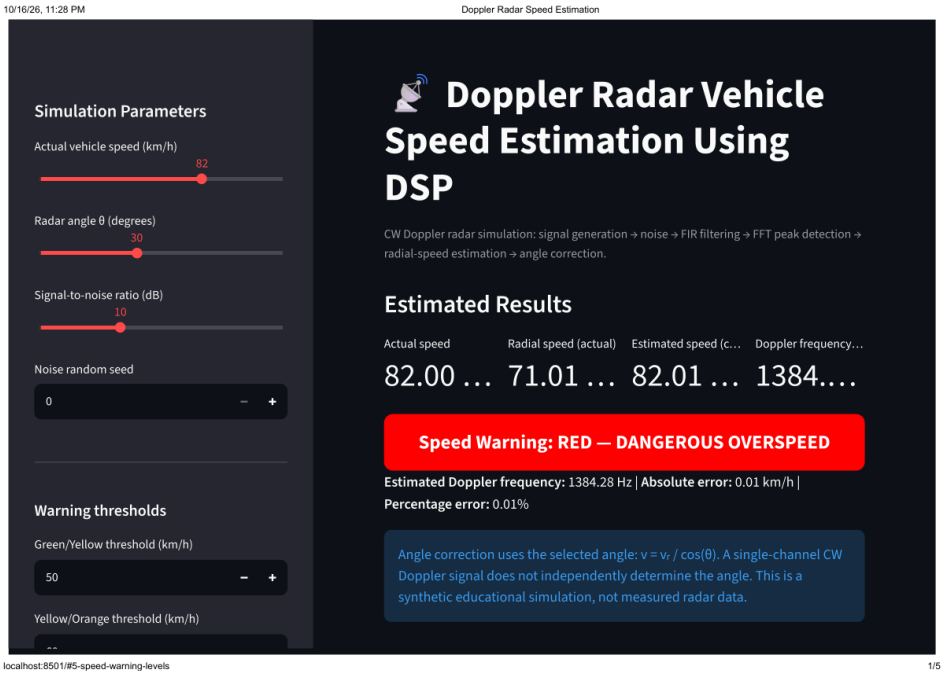


Figure 6.1: Simulation dashboard showing selected parameters, estimated speed, detected Doppler frequency, reported error, and warning level.

6.3 Time-Domain and Frequency-Domain Observations

The time-domain plots show the synthetic signal and the effect of noise and filtering. The FFT spectrum presents the dominant frequency component used for the speed calculation. The actual-versus-estimated plot compares the simulation input with the calculated output. These displays make it possible to inspect the signal-processing stages visually.

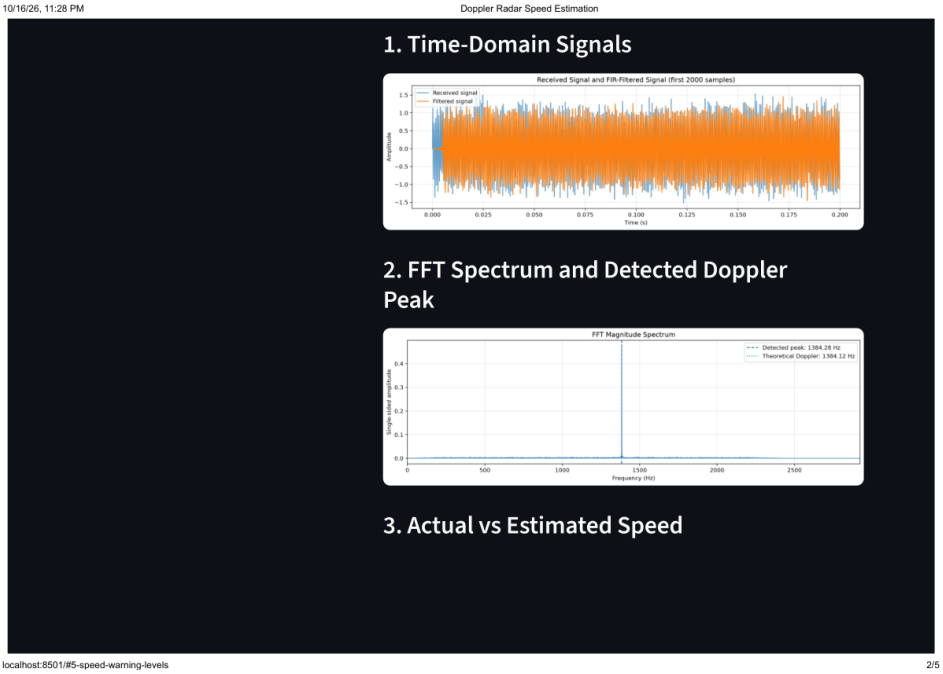


Figure 6.2: Time-domain signals, FFT spectrum with detected peak, and simulated actual-versus-estimated speed.

6.4 Angle Effect and Warning Levels

The angle plot illustrates the geometric relationship between path speed and radial speed. As the angle between the radar line of sight and the direction of travel increases, the measured radial component decreases for a fixed path speed. Cosine correction compensates for this effect when the angle is known accurately. The warning plot shows the category selected from the configured thresholds.

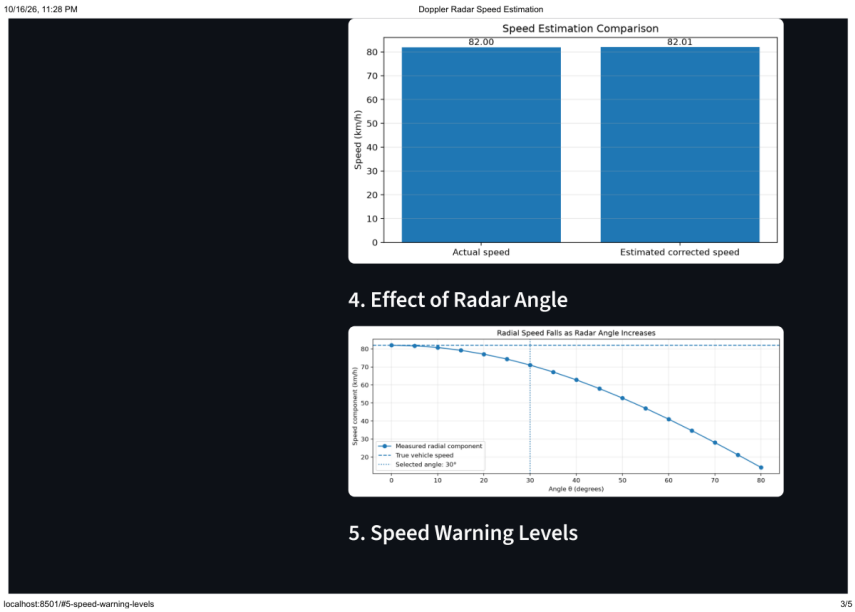


Figure 6.3: Simulated radar-angle effect and speed-warning levels.

6.5 DSP Configuration Details

The configuration panel lists the carrier frequency, wavelength, sampling frequency, number of samples, FFT length, signal-to-noise ratio, FIR order, and Doppler-frequency limits. The detected peak differs slightly from the theoretical Doppler frequency. Finite sampling, FFT-bin spacing, filtering, and added noise can contribute to this difference.

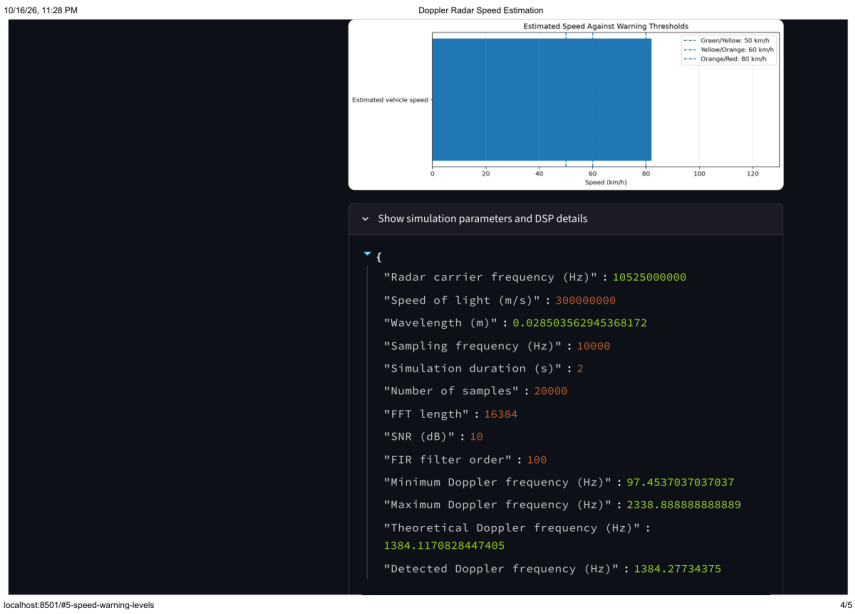


Figure 6.4: DSP parameters and theoretical and detected Doppler frequencies reported by the simulation.

6.6 Discussion and Observation

For the documented run, the theoretical Doppler frequency is 1384.117 Hz and the detected frequency is 1384.277 Hz. The simulation reports an estimated speed of 82.01 km/h against a selected input of 82.00 km/h, with a displayed absolute error of 0.01 km/h and percentage error of 0.01%. The result indicates that the processing chain recovers the configured synthetic input closely under this set of simulation conditions.

Observation boundary: The small reported error is calculated against a known simulated input. It is not a measured error against a real moving vehicle and does not establish the accuracy of a physical radar system. Additional simulation trials across different speeds, angles, noise levels, and random seeds would be needed for a broader algorithmic evaluation.

CHAPTER 7
ADVANTAGES AND LIMITATIONS

7.1 Advantages

- Demonstrates contactless speed estimation using the Doppler principle.
- Combines signal generation, noise addition, filtering, FFT analysis, and speed conversion in one workflow.
- Allows parameters to be changed for repeatable simulation trials.
- Uses Hamming-window FIR filtering to reduce unwanted frequency components before spectral analysis.
- Illustrates the distinction between radial velocity and vehicle-path speed.
- Provides numerical output and visual plots for educational analysis.

7.2 Limitations

- The project is a simulation only; no physical radar sensor or hardware data acquisition was implemented.
- The synthetic signal may not represent all characteristics of real radar returns, road clutter, multipath, or interference.
- A single-channel CW signal does not independently determine target range or radar deployment angle.
- A dominant-peak detector may be unreliable when multiple targets or strong interfering components are present.
- The displayed error is not a validated real-world accuracy claim.
- The simulation does not establish real-time performance on an embedded DSP platform.

CHAPTER 8
FUTURE SCOPE

The simulation can be extended to provide a more comprehensive evaluation of Doppler-based speed estimation. Possible improvements include:

- Run repeated trials over a wider range of speeds, radar angles, signal-to-noise ratios, and noise seeds, and report mean and maximum error.
- Compare different window functions and FIR filter designs to study their effect on spectral peak detection.
- Investigate interpolation around the FFT peak to improve frequency estimation beyond the nearest FFT bin.
- Extend the model to include multiple targets, clutter, and more realistic interference.
- Add direction estimation using quadrature in-phase and quadrature channels.
- Compare the simulation output with recorded radar data if suitable equipment becomes available.
- Only after algorithm validation, consider implementation on a microcontroller or embedded DSP platform with suitable sampling hardware.

CHAPTER 9
CONCLUSION

A software simulation of Doppler radar vehicle speed estimation was developed to demonstrate the application of digital signal processing to a radar problem. The simulated processing chain generates a Doppler signal, adds noise, applies a Hamming-window FIR low-pass filter, detects the dominant spectral peak using the FFT, converts the frequency to radial velocity, and applies radar-angle correction. A warning category is then selected using configurable speed thresholds.

In the documented simulation observation, the detected Doppler frequency is close to the theoretical value, and the estimated speed is close to the selected synthetic input. The plots provide a visual view of the time-domain signal, frequency spectrum, speed comparison, angle effect, and warning classification. Since the project was implemented only in simulation, the results demonstrate the operation of the algorithm under controlled conditions and must not be treated as physical radar measurements or proof of real-world accuracy. Future work can evaluate performance across repeated simulations and, if hardware becomes available, validate the method using measured radar signals.

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